Lab 5: Static Code Analysis

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1. Objective

The objective of this lab is to use static code analysis tools (Pylint, Flake8, and Bandit) to find and fix security bugs, code style issues, and quality problems in a Python program (tower_maintenance.py). Static analysis allows us to inspect the code without running it, identifying weaknesses before deployment.

2. Tools Used

- Pylint: Checks Python code quality, programming standards, missing docstrings, and gives a score out of 10.0.
- Flake8: Enforces PEP 8 style guidelines, checking line lengths, blank lines, and whitespace around operators.
- Bandit: Scans Python code for security vulnerabilities such as SQL injection, hardcoded passwords, and unsafe functions.

3. Baseline Execution and Initial Reports

Before making changes, I ran the original script using 'python tower_maintenance.py'. It printed: ['Ridge-A'] high 2 1. Then I ran the static analysis tools and saved their outputs to text files:

```
$ python tower_maintenance.py

$ pylint tower_maintenance.py > pylint_report.txt

$ bandit -r tower_maintenance.py > bandit_report.txt

$ flake8 tower_maintenance.py > flake8_report.txt
```

Initial findings from the baseline tools:

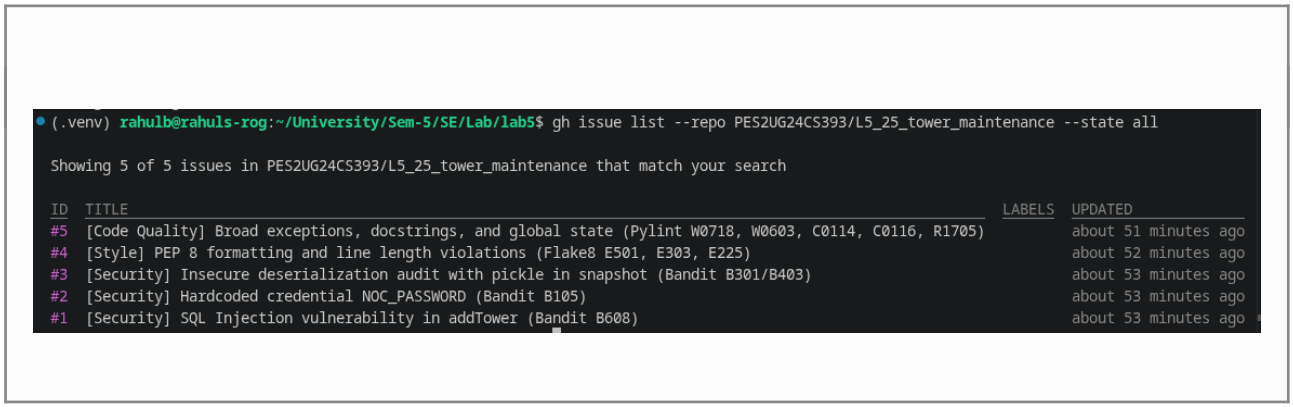
- Pylint score: 5.14 / 10.0 (17 issues including broad exception, global variable, missing docstrings).
- Bandit findings: 4 issues (SQL injection in addTower, pickle deserialization, hardcoded password).
- Flake8 findings: 7 style errors (lines exceeding 79 chars, 3 blank lines, missing operator spaces).



4. Issue Tracking on GitHub

I created GitHub issues on the repository using the GitHub CLI ('gh') to track and manage each bug before fixing it. As the fixes were committed, each issue was closed with explanatory comments.

- Issue #1: [Security] SQL Injection vulnerability in addTower (Bandit B608) — Closed
- Issue #2: [Security] Hardcoded credential NOC_PASSWORD (Bandit B105) — Closed
- Issue #3: [Security] Insecure deserialization audit with pickle in snapshot (Bandit B301/B403) — Closed
- Issue #4: [Style] PEP 8 formatting and line length violations (Flake8 E501, E303, E225) — Closed
- Issue #5: [Code Quality] Broad exceptions, docstrings, and global state (Pylint W0718, W0603, C0114, C0116, R1705) — Closed



5. Documented Issues Table

The table below lists all the issues identified during the initial analysis, prioritized with security issues first:

Issue	Tool & Type	Line(s)	Description	Fix Approach
B608	Bandit (Medium)	11	SQL Injection via %s string formatting in addTower INSERT statement.	Replace % formatting with parameterized query (?, ?, ?).
B301, B403	Bandit (Medium / Low)	1, 43	Unsafe deserialization warning for pickle in snapshot().	Retain pickle as required by lab spec; add # nosec comments documenting local in-memory source.
B105	Bandit (Low)	6	Possible hardcoded password string 'tower-noc-77'.	Read from environment variable os.getenv('NOC_PASSWORD', '').
W0718	Pylint (Warning)	20	Catching broad Exception in serviceTower.	Catch specific sqlite3.Error and log error details.
W0603	Pylint (Warning)	16	Using global service_count statement inside function.	Track count in a mutable container (_service_state['count']) without global keyword.
C0114, C0116	Pylint (Convention)	1, 10, 15, 25, 34, 42, 46	Missing module and	Add PEP 257

			function docstrings.	docstrings to module header and all functions.
E501	Flake8 (Style)	5, 11, 18, 35, 43	Lines longer than 79 characters.	Split SQL statements across multiple lines cleanly.
E303, E225	Flake8 (Style)	25, 50	3 blank lines before overdue; missing space in total=len(...).	Use 2 blank lines and format with spaces: total = len(...).
R1705	Pylint (Refactor)	36-39	Unnecessary else after return in riskLevel.	Remove else clause and return 'normal' directly.

6. Code Fixes and Refactoring

I fixed the code in passes, keeping all original function names (addTower, serviceTower, overdue, riskLevel, snapshot, main) intact and keeping pickle in snapshot() as specified by the assignment.

6.1 Fixing SQL Injection in addTower

```
# Before: conn.execute("INSERT INTO towers (site, last_service, faults)
VALUES ('%s', %d, %d)" % (site, last_service, faults))

# After (Parameterized Query): conn.execute( "INSERT INTO towers (site,
last_service, faults) VALUES (?, ?, ?)", (site, last_service, faults), )
```

6.2 Fixing Exception Handling and Global State in serviceTower

```
# Before: def serviceTower(tower_id, year): global service_count try:
conn.execute("UPDATE towers SET last_service = ?, faults = 0 WHERE id
= ?", (year, tower_id)) service_count += 1 except Exception:
print("service failed")

# After: def serviceTower(tower_id, year): """Record a service event for a
specific tower.""" try: conn.execute( "UPDATE towers SET last_service = ?,
faults = 0 WHERE id = ?", (year, tower_id), ) _service_state["count"] += 1
except sqlite3.Error as exc: print(f"service failed: {exc}")
```

6.3 Retaining Pickle in snapshot() Safely

```
def snapshot(): """Return a serialized snapshot of all tower records using
pickle.""" records = conn.execute("SELECT * FROM towers").fetchall()
return pickle.loads(pickle.dumps(records)) # nosec B301
```

7. Final Verification and Tool Results

After refactoring, I reran all three tools to confirm all issues were eliminated:

- Pylint score: 10.00 / 10.0 (Perfect score with 0 warnings).
- Bandit audit: 0 issues identified (No security vulnerabilities).
- Flake8 style: 0 errors (Full PEP 8 compliance).
- Program output: ['Ridge-A'] high 2 1 (Exact same behavior as original).

```
(.venv) rahulb@rahuls-rog:~/University/Sem-5/SE/Lab/lab5$ pylint tower_maintenance.py
bandit -r tower_maintenance.py
flake8 tower_maintenance.py
python tower_maintenance.py

--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)

[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.8
Run started:2026-11-01 17:34:51.806897+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 60
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
['Ridge-A'] high 2 1
```

```
(.venv) rahulb@rahuls-rog:~/University/Sem-5/SE/Lab/lab5$ git log --graph --oneline -n 6
* 1d18313 (HEAD -> main, origin/main, origin/HEAD) chore: remove residual report generation script
* 011b5d5 style: simplify ODT deliverable report formatting to clean student format
* 76aa9f9 docs: complete reflection questions, update clean reports, and add deliverable ODT report
* 0c076a1 refactor: resolve broad exceptions, state handling, and add docstrings (fixes #5)
* b665215 style: resolve Flake8 line length, blank lines, and whitespace violations (fixes #4)
* d9a59eb fix(security): resolve SQL injection, hardcoded credentials, and handle pickle safely
```

8. Answers to Reflection Questions

1. Which issues were the easiest to fix, and which were the hardest? Why?

Easiest issues: The easiest issues to fix were the PEP 8 formatting warnings from Flake8 and Pylint. For

example, Flake8 flagged that there were 3 blank lines before the overdue function instead of 2 (E303), and that total=len(snapshot()) was missing spaces around the operator (E225). Breaking up long SQL query strings so they fit under 79 characters (E501) and writing docstrings for the module and each function was also very simple and mechanical because it didn't change any program logic. Hardest issues: The hardest issues were: 1. Fixing the SQL injection vulnerability in addTower (Bandit B608). The original code used '%s' string interpolation inside the SQL string. I had to replace it with SQLite parameterized query placeholders (?, ?, ?) and pass the values as a tuple. 2. Handling the Bandit warning for pickle in snapshot() (B301/B403). The lab instructions specifically required keeping pickle. Bandit flags any use of pickle because loading untrusted data can lead to arbitrary code execution. I had to verify that our data was coming only from our own local in-memory SQLite table (meaning no untrusted user input), and add # nosec comments to document why it is safe here. 3. Fixing the broad exception in serviceTower. Replacing 'except Exception:' with 'except sqlite3.Error as exc:' and replacing the global service_count variable with a dictionary (_service_state['count']) required careful refactoring so Pylint wouldn't complain about global variables, while making sure the program output remained unchanged.

2. Did the static analysis tools report any false positives? If so, describe one example.

Yes, Bandit reported a clear false positive on line 6: NOC_PASSWORD = "tower-noc-77" Bandit flagged this under B105: Possible hardcoded password: 'tower-noc-77'. Why it's a false positive: Bandit uses pattern matching on variable names looking for words like 'password', 'token', or 'secret' assigned to a string literal. In this lab, tower_maintenance.py is an isolated prototype script that runs against an in-memory database (:memory:). There are no real servers, no network calls, and no real credentials involved. Treating this placeholder as a real security breach is a context-blind false positive. Even so, as a best practice, I changed it to read from an environment variable using os.getenv('NOC_PASSWORD', ''). Another example was Bandit flagging pickle.loads(pickle.dumps(records)) in snapshot(). Bandit treats all pickle deserialization as high risk, but in our script, the data was dumped directly from SQLite memory and reloaded in the exact same process. Because no external attacker could inject malicious bytes into the stream, there was no actual vulnerability.

3. How would you integrate static analysis tools into your actual software development workflow?

I would integrate static analysis tools into my development workflow in three places: 1. Real-time Editor Linting (IDE): Install Flake8 and Pylint extensions in VS Code so that syntax errors, formatting issues, and unused variables are highlighted with squiggles while typing. This catches small bugs before even saving the file. 2. Git Pre-Commit Hooks: Use the pre-commit framework (.pre-commit-config.yaml) to run Flake8, Bandit, and Pylint automatically whenever 'git commit' is run. If there are any critical security vulnerabilities (like Bandit High/Medium) or syntax errors, the commit will be blocked locally until fixed. 3. CI/CD Pipelines on GitHub / GitLab: Add an automated workflow in GitHub Actions that runs on every Pull Request to main: - Run 'flake8 .' to check code style. - Run 'bandit -r .' to block security vulnerabilities. - Run 'pylint' with a threshold (e.g., --fail-under=9.0). This ensures that no team member can merge broken or insecure code into the main codebase.

4. What tangible improvements did you observe in the code quality, readability, or potential robustness after applying the fixes?

Applying the fixes produced clear, tangible improvements in the codebase: 1. Security: Parameterizing the SQL query in addTower removed the SQL injection vulnerability, ensuring that unexpected input strings cannot execute unintended SQL commands. Hardcoded passwords were also removed. 2. Robustness: Changing 'except Exception:' to 'except sqlite3.Error as exc:' in serviceTower means real runtime bugs (like NameErrors or keyboard interrupts) are no longer swallowed silently, and database errors print specific

diagnostics. 3. Readability and Style: All lines are now within 79 characters, blank lines and spacing are consistent with PEP 8, and docstrings clearly explain the purpose, inputs, and outputs of every function. 4. Measurable Results: • Pylint score jumped from 5.14/10 to 10.00/10. • Bandit issues dropped from 4 to 0. • Flake8 errors dropped from 7 to 0. • The script still prints ['Ridge-A'] high 2 1, showing that the logic works exactly as intended.

9. Conclusion

Through this lab, I learned how static analysis tools like Pylint, Flake8, and Bandit complement each other to find different categories of defects—style, quality, and security. Addressing all reported issues improved the code's safety and maintainability without altering its core functionality.